Angle-Dependent Magneto-Resistance near the Pre-Martensitic Phase of Ni_2MnGa

C. P. Opeil, [a] J. C. Lashley, [b] J. L. Smith [b]

[a] *Department of Physics, Boston College, Chestnut Hill, MA USA*
[b] *Los Alamos National Laboratory, Los Alamos, NM, USA*

Measurements: Isothermal Magneto-Resistance in Fields up to 1 Tesla in the Pre-Martensitic Phase (190 ≤ T ≤ 235 K), T_C = 381 K, T_M = 194 K.

Sample: High quality Ni_2MnGa single crystal grown by Bridgman technique. Characterized by specific heat, resistivity, ARPES, Laue diffraction, thermal expansion and magnetic-moment measurements.

Motivation: Softening of the TA_2 phonon branch in Ni_2MnGa is associated with Fermi surface nesting[Lee] and recent ARPES data show a significant depletion of states (pseudo-gap) forming at 0.3 eV below with Fermi energy[Opeil]. In what way do these effects help us to understand the positive to negative (slope) magneto-resistant switching in this material?

(1) Introduction:

(a) Isothermal magneto-resistance (MR) measurements detect a change in slope from positive to negative at the T_{PM} that can be suppressed with a modest field.

(b) Ni_2MnGa magnet moments are localized on the Mn sites, and result from an exclusion of minority-spin electrons of Mn 3*d* states and strong interactions with Ni *d* conduction electrons.[Barman]

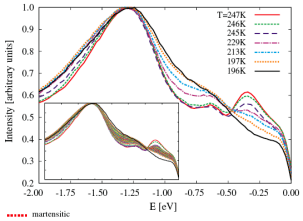
(c) Recent UV photoemission spectra (UVPS) show a pseudo-gap opening at 0.3 eV below the Fermi energy in the pre-martensite temperature (T_{PM}) phase.[Opeil]

(d) Above T_M a peak in UVPS at 0.3 eV is due to Ni *d* spin-down electrons. Below T_M this peak disappears and enhanced DOS appear at 0.8 eV.

(e) First principle electronic structure calculations confirm experimental evidence of Ni *d* and Mn *d* contributions to spin-up DOS.

(2) Pseudo-gap formation, DOS in Ni, Mn

Figure 1. Energy dependence of the normal photoemission intensity for selected temperatures in the region of the PMT in Ni_2MnGa. Lower inset shows *all* intensity plots at 1 K temperature intervals.



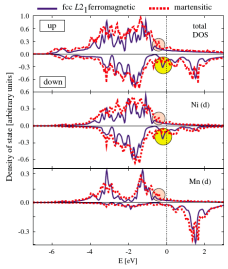


Figure 2. The majority- and minority-spin DOS for the spin-polarized martensitic phase ($T < T_M$) and the spin polarized fcc ($T_M < T < T_C$) of Ni_2MnGa. The top panel depicts the total DOS, whereas the middle and bottom panels depict the contributions due to the *d* electrons of the Ni and Mn atoms, respectively.

C. P. Opeil *et al.*, Phys. Rev. Lett. 100, 165703 (2008).

(3) Resistivity and Isothermal MR

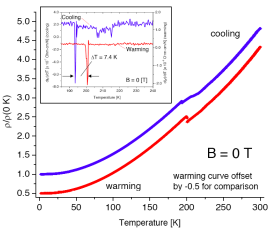
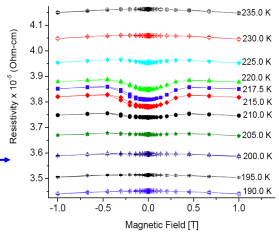


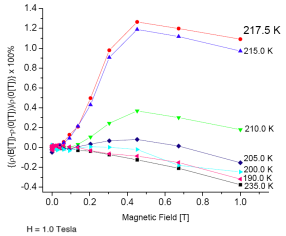
Figure 3. Residual resistivity ratio, as a function of temperature for Ni_2MnGa with applied magnetic field of B = 0 T. Upper curve is the resistivity ratio for data taken upon cooling. Bottom curve illustrates data taken during sample warming and is offset from cooling curve by -0.5 for comparison. The insert shows dρ/dT versus temperature for the martensite, T_M~194 K and pre-martensite (T_{PM} ~194 - 235 K) regions. The thermal hysteresis between martensitic transition temperatures for cooling/warming curves is T ~ 7.4 K.

Figure 4. Isothermal magneto-resistance of Ni_2MnGa as a function of applied magnetic field at temperatures indicated to the right. Between each temperature measurement the sample was heated to 300 K and cooled in zero field. Solid lines are a guide to the eye.



(4) MR % change and angle dependence

Figure 5. The percent change of isothermal magneto-resistance versus applied magnetic field for Ni_2MnGa. Results are symmetric about the vertical axis, however, for clarity only select temperatures and positive values of field are plotted. The largest percent change occurs at 217.5 K (filled circles).



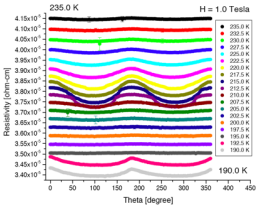


Figure 6. Isothermal magneto-resistance as a function of rotation angle theta from field incident on the [001] plane. Maximum deviation occurs around 215 K.

(5) Discussion of magneto-resistance

(1) Resistance and MR are measures of electron-electron, electron-ion or phonon scattering. Changes in resistance or MR can be thought of as scattering from one band state to another and the energy added to electrons by the applied field.

(2) The switching from "+" to "-" slope in MR for Ni_2MnGa is greatest at T ~ 217.5 K during the pre-martensite "intermediate" phase. This phase is described by a Landau-type model (weakly first-order) which originates from a magneto-elastic coupling between magnetization and the anomalous softening of the TA_2 [110] phonon branch.[Planes]

(3) The pseudo-gap associated with the intermediate phase indicates DOS changes that influence both valence and conduction (Ni) electrons. Electronic structure calculations support the experimental evidence for band structure changes before the martensite transition.

(6) Conclusions

(1) Local magnetism relies on strong interaction of Mn 3*d* states and Ni conduction electrons, one expects a deviation of MR behavior as the pseudo-gap is approached that relies on a redistribution of Mn and Ni *d* states.

(2) The application of a **B** ≥ 0.5 T raises the electrons to an energy sufficient to allow a redistribution of *d* states similar to a condition prior to pseudo-gap onset and thus a return to a negative slope in MR.

(3) The onset of a pseudo-gap provides further evidence that the Fermi surface is strongly nested[Lee] at the MT while it is partially nested at the premartensitic transition (PMT).

(7) References

[1] Y. Lee, J. Y. Rhee and B. N. Harmon, Phys. Rev. B 66, 054424 (2002).

[2] C. P. Opeil *et al.*, Phys. Rev. Lett. 100, 165703 (2008).

[3] A. Planes *et al.*, Phys. Rev. Lett. 79, 3926 (1997).

[4] S. R. Barman et al., Phys. Rev. B **72**, 184410 (2005).

Funding: Trustees of Boston College and the Department of Energy at Los Alamos National Laboratory.